

Matter's implementation of CMAF Ingest allows for optional end-to-end media encryption using the standard procedure defined in [CENCISOBMFF](#) and [CENC](#). [CENC](#) is an ISO standard for supporting end-to-end encryption of CMAF media files. Controllers/Ecosystems MAY utilize CENC in their deployments, but management and generation of the CENC key is outside the scope of this standard. Matter supports both initial CENC key assignment using the [CMAFContainerOptionsStruct](#) struct via the [AllocatePushTransport](#) command and CENC key-rotation by a Controller after initial provisioning using the [ModifyPushTransport](#) command. When CENC mode is active, Nodes SHALL use the AES-128 CBC [cbcs](#) mode of CENC as defined in Section 10.4 of [ISO-23001:7-2023](#) or later.

In CMAF, the basic units of the protocol are Segments, Tracks, and Sessions.

11.7.1.1.1. Segment

A Segment is a fixed duration [ISOBMFF](#) file which contains the Audio, Video, and/or Metadata samples themselves. A Segment MAY multiplex Audio and Video if configured to do so, otherwise the Segment would contain a single type of media.

11.7.1.1.2. Track

A Track is a symbolic name applied to a sequence of one or more [Segments](#), of the same media type, from the same source. In Matter, an [Audio Stream](#) maps to an audio Track, a [Video Stream](#) maps to a video Track, and one or more [metadata content](#) maps to a metadata Track. All Segments within the same Track utilize the same fixed duration, except the last Segment which may be less than or equal to that duration. Additionally, all Segments within the same Track are sequentially numbered per the [Segment Numbering](#) rules.

11.7.1.1.3. Session

A Session is defined as the set of one or more [Tracks](#), that lasts for a duration of time. The duration of a Session can be variable or fixed. The Session's duration is based on the duration of the [Trigger Type](#) that caused its transmission to occur. In Matter's CMAF usage, a Session can be considered a low level logical event, though one delineated only by Motion detection, Command invocation timings, or fixed durations.

11.7.1.2. Operation

CMAF at its core is a sequence of HTTP PUT/POST operations for each media [Segment](#), within each [Track](#), within each [Session](#). Every operation is sent to a unique URL, which is known as the [POST_URL](#) [https://dashif.org/Ingest/#post_url]. Each unique URL contains information about the Session, Track, and Segment. Each unique URL is comprised of two components that are concatenated together.

The first component of the [POST_URL](#) is a static value called the [publishing_point_URL](#) [https://dashif.org/Ingest/#publishing_point_url]. The [publishing_point_URL](#) contains the scheme, host-name of the server, and whatever initial URL path components the receiving-entity would like to receive. Typically, deployments encode user specific information or other ecosystem specific items within the initial URL path values in the [publishing_point_URL](#).

For Matter deployments, the [publishing_point_URL](#) SHALL use the [https](#) scheme, SHALL NOT contain the URL fragment character <#>, SHALL NOT contain the URL query character [?](#), and SHALL end